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Modified Dry Direct Seeding Technology

Rice Technology Bulletin Series

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FOREWORD

The Dry Direct Seeding Technology for rice is a traditional method of rice establishment especially in the rainfed and upland areas in the country. The method was not well adopted because of the occurence of weeds, desiccation of the seeds, more seeds were used and relatively lower yield.

During the past three years, PhilRice through our Branch Station in Batac in collaboration with some farmers and LGU technicians modified the dry direct seeding technique. We now establish in moist soil with pre-germinated seeds instead of dry seeds and dry soil, broadcast over well-pulverized "lithao" prepared furrows. Herbicides to control weeds are now part of the technology.

We found the MDDS technique appropriate in rainfed areas. Our farmer cooperators in Region 1 and Abra are now harvesting as high as 7 t/hectare.

With this technology bulletin, we are giving farmers in the rainfed areas an opportunity to increase prodcutivity in their farms as well as increase their income.

LEOCADIO S. SEBASTIAN
Executive Director

BACKGROUND

Thirty percent of the total rice area in the world depends on rain for irrgation. In Asia alone, 47% of rice fields are rainfed. Of these, about two million hectares are found in the Philippines.



The most common method of rice establishment in the Philippines is transplanting. However, before a farmer can transplant the seedlings, he needs to accumulate enough water, which is approximately 1 million gallons per hectare, for puddling. This amount of water is available only during the months of July and August when the rainy season is at its peak.

If farmers in the rainfed areas wait for the rains to be able to transplant, their plants will be more exposed to typhoons and drought. Delayed transplanting owing to insufficient rain results in lesser tillers, hence, less production and less income.

With the use of the modified dry direct seeding technique (MDDST), farmers in rainfed areas will be able to plant on time and harvest in the early weeks of September without depending so much on rain water. This will enable them to plant other crops earlier in order to use residual moisture.

Problems addressed

- lack of irrigation/erratic water supply
- low crop intensity
- high cost of transplanting
- low income

Outstanding features

- early land preparation and seed establishment
- increases cropping intensity by accomodating another crop after rice
- optimizes residual soil moisture after rice
- maximizes utilization of rainfall/irrigation water
- easier land preparation for upland crop production because the soil structure is not destroyed
- increases income from additional crops

Advantages of Dry Direct Seeding

- saves about PhP3500 per hectare on seedbed preparation, pulling and transplanting costs.
- solves problem on labor, which is scarce during peak of transplanting.
- increases the productivity of the rainfed areas through maximum utilization of available soil moisture allowing two to three croppings per year
- increases net income especially in rice production by about PhP16,000 to PhP18,000 per hectare.

Comparison between the MDDST and the direct seeding technology commonly used by farmers

	MDDST	TRADITIONAL DIRECT SEEDING
Time of sowing	· When the upper 25-30 cm portion of the soil is moist	· Even without adequate soil moisture
Germination	· Seeds are pre-germinated *(In areas where rains can not be predicted and the soil is dry, dry seeds can be sown uniformly over well-pulverized "lithao" prepared furrows)*	· Seeds are not pre-germinated
Establishment	· Seeds are broadcast over a well-pulverized "lithao" prepared furrow	· Seeds are either drilled or dropped along furrows
Amount of seeds	· 60-80 kg/ha of certified seeds	· 120 kg/ha or more seeds
Covering the seeds	· Seeds are covered with a wooden plank, drawn by a carabao, cow, or small hand tractor	· Seeds are covered by a spiked-tooth harrow



The MDDS rice (left) is ready for harvest while the transplanted rice (right) are still at the vegetative stage.

Recommended varieties



PSB Rc14 and PSB Rc12 are appropriate for dry direct seeding. PSB Rc14 matures within 114 days while PSB Rc12 matures within 110 days. Both varieties are moderately tolerant to drought. The potential yield of both varieties is 6t/ha but it could reach as high 9t/ha.

Land preparation

Prepare the area to be planted immediately after harvesting the dry season crop. The field should be plowed once or twice and harrowed two to three times to have a well-pulverized soil. Thorough land preparation will expose weed seeds to the sun, hence, desiccating them. This reduces weed population. A well-pulverized soil also promotes better germination of the seeds.

To test if the soil is ready for seeding put some soil in one palm and squeeze it. If the soil does not compact in the hands, then it is ready for sowing.

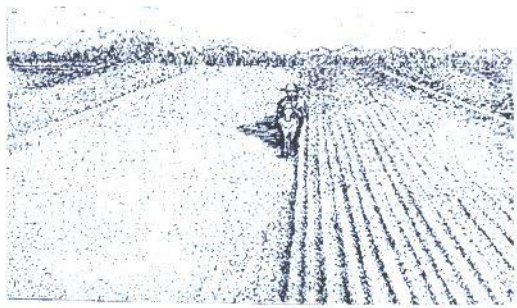


Basal fertilizer application

- For best results, incorporate the recommended basal fertilizer based on the results of soil analysis during final harrowing.
- In the absence of soil analysis, the general recommendation of 10 sacks of organic fertilizer (50 kg/sack) and 20-30 kg NPK/ha (3-4 bags 14-14-14) can be applied. This will provide the nutrients needed by the plants during the early growth stages. Cover the fertilizers to avoid nutrient losses.

Making the furrows

- After covering the fertilizer, make the furrows at 20 cm apart.
- Use a "lithao" or a plow without the moldboard to make the furrows.



Preparing the seeds to be sown

- Prepare 60-80 kg of seeds for a hectare.
- If possible wash the seeds to allow unfilled grains to float. Remove seeds that float.

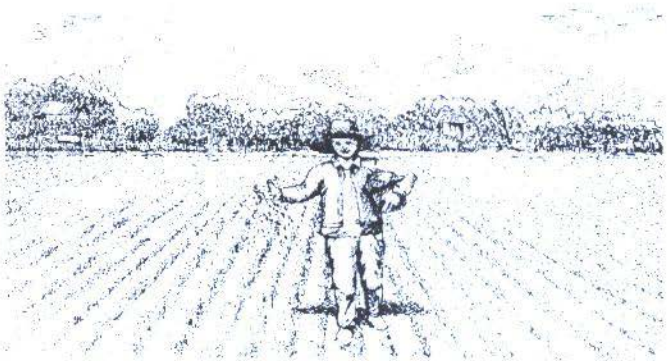




- Provide enough space in the sack for seed expansion.
- Soak seeds within 8-12 hours and incubate 8-12 hours also before sowing.
- The seeds will be ready for sowing the moment a whitish outgrowth appears at the base of the seeds.

Sowing

- Sow the seeds immediately after making the furrows, preferably early morning or late in the afternoon, when the upper 25-30 cm portion of the soil is already moist. This amount of moisture is enough to sustain the germination of the seeds.



- Broadcast the seeds uniformly. Cover seeds immediately by passing a wooden plank drawn diagonally by a farm animal to the direction of the furrows. This will put seeds that fell on top of the ridges inside the furrows.



Weed control



Weeds are a major problem in direct-seeded fields. However, this problem can be remedied by thorough land preparation, use of herbicides, spot weeding, and efficient water management.

- Apply pre-emergence herbicide 1-2 days after broadcasting and cover the seeds provided that the soil is moist.
- Apply early post emergence herbicide up to 8 days after sowing. This herbicide controls broad leaf weeds, sedges, and grasses. Post emergence herbicides can be applied at 25-30 days after emergence.

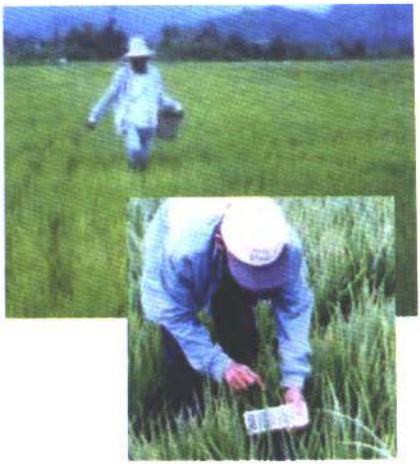
Water management

Fix dikes and outlets of water in paddies to trap all rainwater available. Maintaining water in the paddies also help control weeds.



Fertilizer management

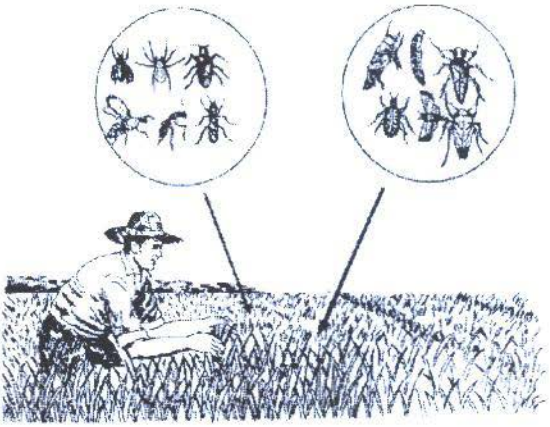
- Topdress 15-25 days after seed emergence. Apply 2 bags of urea per hectare. Succeeding application of nitrogenous fertilizers should be based on the Leaf Color Chart (LCC).



Pest management

Monitor the field regularly. Use the integrated pest management (IPM) approach.

- Practice synchronous planting in the community to reduce pests in the field.
- Use pesticides efficiently and cautiously to maintain beneficial organisms in the field.



Harvesting

Harvest the plants when 80 percent of the grains in the panicles have turned golden yellow.

Result of MDDS trial in different locations, WS 2000-2001.

Location	Yield	Production Cost	Gross income	Net income
Batac, Ilocos Norte	4,673	22,857	56,076	33,219
Sta. Lucia, Ilocos Sur	5,993	18,787	55,778	36,991
Bacnotan, La Union	7,000	17,858	56,636	38,778
Pidigan, Abra	4,855	18,711	41,267	22,556

FARMER-COOPERATORS

Pidigan, Abra
- Francisco Cabaya Jr.
- Vicente Palino
- Conrado Lajon
- Alejandro Cabaya Jr.
- Nelson Felicitas
- Simon Bisquera
- Artemio Urbano
- Pedro Apolinar
- Franklyn uanang
- Carlito Pada
- Marcelino Bisquera
- Soledad Berja
- Juan Basingan
- Santos Beleno
- Florencio Bisquera
- Saturnino Lajon
- Aurencio Palino
- Alfredo Corpuz Jr.
- Ricky Sanidad
- Constancia Padapat
- George Ely Paz
- Amor Beleno
- Jose Pablico
- Eugenio Madriaga
- Reynalldo Madriaga
- Allan Pada
- Danny Blancaflor
- Alejandro Bersalona

Danglas, Abra
- Samuel Ballestraa
- Rogelio Masaoay

Aringay, La Union
- Mario Mangaoang

Agoo, La Union
- Ceasaar Mendoza

Tubao, La Union
- Nelson Padilla

Bauang, La Union
- Rogelio Ramos

Bacnotan, La Union
- Chinita Pascua

Batac, Ilocos Norte
- Armel Alcantara
- Mariano Agloco Sr.
- Rafael Malubay
- William Samson
- Mariano Agloco Sr.
- Rey Manglal-lan
- Luciano Manglal-lan
- Miguel Tulali

Currimao, Ilocos Norte
- Alfredo Allado

San Juan, Ilocos Sur
- Larry Arocena
- Guillermo Varilla

Sta Lucia, Ilocos Sur
- Jose Valdez
- Demetrio Marquez
- David Corbita
- Romeo Corbito

Subject Matter Specialists

Mr. Samuel P. Liboon	*Science Res. Specialist, PhilRice*
Ms. Nida Q. Abrogena	*Research Fellow, PhilRice*
Ms. Alma C. Aguinaldo	*Research Fellow, PhilRice*
Ms. Perpetua B. Ancheta	*Assisting Technologist, Naguirayan, Pidigan, Abra*
Ms. Marlita B. Basingan	*Municipal Agriculturist, Pidigan, Abra*
Mr. Jose Valdez	*Assisting Technologist, Sta. Lucia, Ilocos Sur*
Dr. Expedito Villaruz	*Assisting Technologist, San Diego, Pidigan, Abra*
Ms. Teresita Eslava	*Assisting Technologist, Bacnotan, La Union*
Engr. Nelson Diculen	*Assisting Technologist, Magnuang Batac, Ilocos N*
Dr. Romeo Lucero	*Municipal Agriculturist, Danglas, Abra*

Managing Editor/Desktop Artist
Antoinette A. Kindipan

Cover Design
Carlo G. Dacumos

Illustrations
Carlito N. Bibal

Editorial Advisers
Reynaldo C. Castro
Leocadio S. Sebastian
Karen Eloisa T. Barroga

For further information, contact:

Philippine Rice Research Institute
Batac, 2906 Ilocos Norte
Tel. No. (077) 792-2545; Telefax (077) 792-4702

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PhilRice

PhilRice is a government-owned and -controlled corporation attached to the Department of Agriculture. It was created through Executive Order 1061 dated 5 November 1985 (as amended) to help develop high-yielding and cost-reducing technologies so farmers can produce enough rice for all Filipinos.

It accomplishes this mission through reserach, development and extension (RD&E) through its central and branch stations coordinating with a network that includes 57 agencies and 95 seed centers strategically located nationwide.

PhilRice is an ISO14001-certified agency.

Its interdisciplinary programs are the following: (1) direct-seeded and (2) transplanted irrigated lowland rice; (3) hybrid rice; (4) rice and rice-based products; (5) rice-based farming systems; (6) policy research and advocacy; and (7) technology promotion. With these programs, PhilRice develops and promotes technologies that are ecosystem-based, location- and problem-specific, and profitable to the Filipino farmers.

for more information, write, visit or call:

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INTERNATIONAL YEAR OF RICE 2004

Department of Agriculture

PHILRICE

Philippine Rice Research Institute

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PhilRice
Central Experiment Station
Science City of Muñoz, 3119 Nueva Ecija
Trunklines: 63 (44) 456-0277, -0258, -0649, -0426, -0433
E-mail: prri@philrice.gov.ph
Website: http://www.philrice.gov.ph

PhilRice *Los Baños*
UPLB Campus, College, 4031 Laguna
Tel/Fax: (49) 536-3631 to 33, 536-3515
E-mail: los_banos@philrice.gov.ph

PhilRice *Isabela*
San Mateo, 3318 Isabela
Tel/Fax: (78) 664-2280, -2954
E-mail: san_mateo@philrice.gov.ph

PhilRice *Batac*
Batac, 2906 Ilocos Norte
Tel/Fax: (77) 792-4714, -4702
E-mail: batac@philrice.gov.ph

PhilRice *Negros*
Cansilayan, Murcia, 6129 Negros Occidental
Tel/Fax: (34) 446-3403
E-mail: negros@philrice.gov.ph

PhilRice *Midsayap*
Bual Norte, Midsayap, 9410 North Cotabato
Tel/Fax: (64) 229-7242, -8178
E-mail: midsayap@philrice.gov.ph

PhilRice *Agusan*
Basilisa, RTRomualdez, 8611 Agusan del Norte
Tel/Fax: (85) 818-2277, -3377, -4477
E-mail: agusan@philrice.gov.ph